Destructuring

1. Object destructuring

You have:

const user = { name: "Amit", age: 25, city: "Delhi" };

Extract name and city and print them.

---

2. Array destructuring

const colors = ["red", "green", "blue", "black"];

Extract first two values into variables and store the remaining in another variable.

---

3. Nested destructuring

```
const user = {
  name: "A",
  address: { city: "Delhi", pincode: 110001 }
};
```

Extract city.

---

4. Destructuring in function parameter

const student = { name: "A", marks: 90 };

Write a function that prints name and marks using destructuring.

Map

5. Add grace marks

const marks = [50, 60, 70, 80];

Add 5 marks to each student using map.

---

6. Convert array to objects

const names = ["A", "B", "C"];

Convert into:

[{name: "A"}, {name: "B"}, {name: "C"}]

---

7. Price update

```
const products = [
  {name: "Shirt", price: 500},
  {name: "Shoes", price: 1500}
];
```

Increase price of each product by 10%.

Filter

8. Passing students

```
const marks = [35, 60, 20, 80, 45];
```

Return only marks greater than or equal to 40.

---

9. Even numbers

From an array, return only even numbers.

---

10. Expensive products

Filter products with price greater than 500.

Find

11. Find user by id

```
const users = [
  {id: 1, name: "A"},
  {id: 2, name: "B"},
  {id: 3, name: "C"}
];
```

Find the user with id = 2.

---

12. Find first match

Find the first number greater than 50 from an array.

Arrow Functions

13. Convert function

Convert:

```
function add(a, b) {
  return a + b;
}
```

into arrow function.

---

14. Implicit return

Write an arrow function that returns square of a number using implicit return.

Combined Scenarios

15. Student processing

```
const students = [
  {name: "A", marks: 80},
  {name: "B", marks: 35},
  {name: "C", marks: 60}
];
```

Tasks:

1. Get students with marks >= 40
2. Increase their marks by 5
3. Return only names

---

16. Cart system

```
const cart = [
  {name: "Shirt", price: 500},
  {name: "Shoes", price: 1500},
  {name: "Cap", price: 200}
];
```

Tasks:

1. Get total price
2. Get products with price > 500

3. Get only product names

---

17. Filter + Map

From an array:

- Filter numbers greater than 10
- Then square them

---

18. Find + Destructuring

Find a user and directly extract their name using destructuring.